

Effective Programming Practices for Economists

Basic Python

Tracebacks

Janoś Gabler and Hans-Martin von Gaudecker

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- What are Exceptions and Tracebacks?
- The Anatomy of a Python Traceback
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Motivation

- We sometimes told you that you cannot do certain things:
 - Example: Can only use hashable objects as dictionary keys
- Now we will discuss what happens if you do it anyways
 - Exception: What class of error?
 - Traceback: Detailed report that helps you to localize the error
- Pro tips
 - Read the traceback!
 - Have it translated!

Example Traceback

```
>>> d = {"a" : 1}
>>> d[[1, 2, 3]] = "b"
>>> d["c"] = 3
```

```
-----
TypeError 1 Traceback (most recent call last)
/home/janos/some_file.py line 2
      1 d = {"a" : 1}
 2----> 2 d[[1, 2, 3]] = "b"
      3 d["c"] = 3
```

```
TypeError: unhashable type: 'list' 3
```

- The code on the left has a problem
- Traceback tells us everything we need:
 1. What type of Exception occurred:
TypeError
 2. Where did it occur: In line 2 of
some_file.py
 3. What happened exactly (*used an unhashable type where we must not*)
- Tracebacks can get very long! Read from bottom to top.
- Always look for these three things!

Common sources of errors

- `ValueError` : Called a function with something invalid
- `KeyError` : Typo in a variable name or a dictionary key
- `TypeError` : Called a function with something that has the wrong type
- `ImportError` : Typo in an import

First step: Ask an AI

- Got a traceback you can't parse? → Paste it into an AI chatbot (Claude, ChatGPT, ...)
- Select and **copy the full traceback text** — not a screenshot
- Add one sentence of context: what you were trying to do

I am working on a Cobb-Douglas function.

Here is the error I got:

[full traceback text]

Can you explain what went wrong?

When AI doesn't resolve it

Try these first:

1. Read the AI's explanation carefully
2. Apply the suggested fix
3. If a new error appears → repeat

Escalate to human help when stuck:

- Ask on the course chat (Zulip course stream — not DM)
- Show: what you tried, the AI's explanation, why it didn't work
- Attach a minimal reproducible example

How not to ask for help

- "I wanted to do the exercise but it does not work"
- "Python does not work on my computer"
- "My code does not work, here is a screenshot"
- Asking via DM on Zulip instead of the course stream
- Sending a screenshot of the error — copy the text instead

A better way

In the task where we should use Python to calculate the output value of a Cobb-Douglas production function (assignment 1, exercise 2) the following line:

```
cobb_douglas(labor, capital, alpha)
```

gives me a type error:

```
TypeError: unsupported operand type(s) for ** or pow(): 'tuple' and 'float'
```

I don't understand the error because I'm just passing in numbers.

A better way (continued)

Here is a minimal example to reproduce the error:

```
labor = 2.5,  
capital = 4.5  
alpha = 0.33  
  
def cobb_douglas(labor, capital, alpha):  
    return labor**alpha * capital**(1 - alpha)  
  
cobb_douglas(labor, capital, alpha)
```

I attach the entire traceback as text. *(copied from the terminal, not a screenshot)*

My favourite AI chatbot suggested the following:

```
...
```

but trying that out did not help and I am stuck.